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On the γ -positivity of multiset Eulerian polynomials

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ABSTRACT

Let $A_n^{(p)}(t)$ (resp. $B_n^{(p)}(t)$) be the descent polynomial on permutations of the multiset $\{1^p, 2^p, \dots, n^p\}$ (resp. $\{1^p, 2^p, \dots, n^p, n+1\}$). The γ -positivity of $A_n^{(p)}(t)$ was known but to give a combinatorial interpretation for the corresponding γ -coefficients still remains open. In this paper we

- manage to find a combinatorial interpretation for the γ -coefficients of $A_n^{(p)}(t)$ via the model of weakly increasing trees;
- and prove that $B_n^{(p)}(t)$ has bi- γ -positivity expansion

$$B_n^{(p)}(t) = a_n(t) + tb_n(t),$$

where $b_n(t) = (p-1)A_n^{(p)}(t)$.

The first result, whose proof makes use of Chen's general bijective algorithm for trees and a new decomposition of weakly increasing trees, answers a recent open problem posed by Lin–Ma–Ma–Zhou. The latter result, which we provide both a computational proof and a short combinatorial proof, extends a bi- γ -positivity result due to Ma–Ma–Yeh from $p=2$ to general p . Their proof for $p=2$ uses Chen's context-free grammar and seems hard to be adopted for general p .

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1. Introduction

Let $M = \{1^{p_1}, 2^{p_2}, \dots, n^{p_n}\}$ be a multiset and $p = p_1 + \dots + p_n$. A multipermutation of M is a word $\pi = \pi_1 \dots \pi_p$ such that k appears p_k times for each $1 \leq k \leq n$. Let $\mathfrak{S}_M$ be the set of all

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multipermutations of M . Let $[n] := \{1, 2, \dots, n\}$. A *descent* (resp. *ascent*) of $\pi \in \mathfrak{S}_M$ is an index $i \in [p-1]$ such that $\pi_i > \pi_{i+1}$ (resp. $\pi_i < \pi_{i+1}$). Denote by $\text{des}(\pi)$ and $\text{asc}(\pi)$ the number of descents and ascents of π , respectively. The *multiset Eulerian polynomial* $A_M(t)$ is defined by

$$A_M(t) = \sum_{\pi \in \mathfrak{S}_M} t^{\text{des}(\pi)} = \sum_{\pi \in \mathfrak{S}_M} t^{\text{asc}(\pi)}.$$

MacMahon [18, Volume 2, p. 211] (see also [13]) proved the factorial generating function formula

$$\frac{A_M(t)}{(1-t)^{p+1}} = \sum_{k \geq 0} \binom{p_1+k}{p_1} \cdots \binom{p_n+k}{p_n} t^k. \quad (1.1)$$

Let $h(t) = \sum_{i=0}^n a_i t^i$ be a polynomial with nonnegative real coefficients. We say that $h(t)$ is *unimodal* if $a_0 \leq a_1 \leq \dots \leq a_{k-1} \leq a_k \geq a_{k+1} \geq \dots \geq a_n$ for some k . Such an index k is called a *mode* of $h(t)$. If $h(t)$ is *symmetric* (i.e., $a_i = a_{n-i}$ for $0 \leq i \leq \lfloor n/2 \rfloor$), then it can be expanded as $h(t) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_k t^k (1+t)^{n-2k}$. When the γ -coefficients $\gamma_k \geq 0$ for $0 \leq k \leq \lfloor n/2 \rfloor$, the polynomial $h(t)$ is called γ -*positive*. Note that the γ -positivity of $h(t)$ implies the unimodality and the symmetry of $h(t)$ and thus $\lfloor n/2 \rfloor$ is a mode of $h(t)$. Brändén [4, Remark 7.3.1] observed that if $h(t)$ is symmetric and has only real roots, then $h(t)$ is γ -positive.

When $M = [n]$, the multiset Eulerian polynomial $A_M(t)$ becomes the classical *nth Eulerian polynomial* (see [21, p. 22]) that we denote $A_n(t)$. Foata and Schützenberger [10] proved in 1970 that $A_n(t)$ is γ -positive and provided a combinatorial interpretation for the corresponding γ -coefficients in terms of permutations without double descents. Since then, several generalizations and analogues of this γ -positivity result have been found; see the two surveys [2,4] and the references therein.

For $p \geq 1$, let $[n]_p := \{1^p, 2^p, \dots, n^p\}$. When $M = [n]_p$, the multiset Eulerian polynomial $A_M(t)$ is called *p-multiset Eulerian polynomial* and will be denoted by $A_n^{(p)}(t)$. In particular, $A_n^{(1)}(t) = A_n(t)$, the *nth Eulerian polynomial*, which is the *h*-polynomials of the dual permutohedra. Recent work of Ardila [1] indicates that $A_n^{(2)}(t)$ is the *h*-polynomials of the dual bipermutohedra, which was first realized in [15,16]. Although $A_M(t)$ is not symmetric in general, Carlitz and Hoggatt [6] proved that $A_n^{(p)}(t)$ is symmetric for any positive integer p . On the other hand, Simion [19] proved that $A_M(t)$ has only real roots for any M . It then follows from Brändén's observation mentioned above that $A_n^{(p)}(t)$ is γ -positive. In [15] Lin, Ma, Ma and Zhou found a combinatorial interpretation for the γ -coefficients of $A_n^{(2)}(t)$ in terms of weakly increasing trees and posed the following open problem.

Problem 1.1 (Lin–Ma–Ma–Zhou [15]). Is there any combinatorial interpretation for the γ -coefficients of $A_n^{(p)}(t)$ for any $p > 2$?

In this paper, we solve this problem by providing an interpretation for γ -coefficients of $A_n^{(p)}(t)$ using the model of weakly increasing trees. Our approach requires the use of Chen's general bijective algorithm for trees [7] and a new decomposition for weakly increasing trees.

Weakly increasing trees were introduced by Lin et al. [15] as unified generalization of plane trees and increasing trees. Recall that a *plane tree* is a rooted tree in which the children of each node are linearly ordered.

Definition 1.2 (Weakly Increasing Trees [15]). Let $M = \{1^{p_1}, 2^{p_2}, \dots, n^{p_n}\}$ be a multiset and $p = p_1 + \dots + p_n$. A *weakly increasing tree* on M is a plane tree such that

- (i) it contains $p+1$ nodes that are labeled by elements from the multiset $M \cup \{0\}$,
- (ii) the root is labeled by 0 and each node receives a label weakly greater than its parent,
- (iii) the labels of the children of each node are weakly increasing from left to right.

See Fig. 1 for an example of weakly increasing tree. Denote by $\mathcal{T}_M$ the set of weakly increasing trees on M .

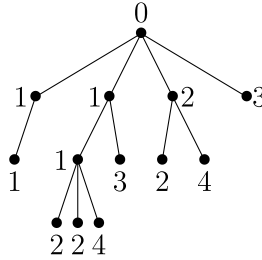


Fig. 1. A weakly increasing tree on $M = \{1^4, 2^4, 3^2, 4^2\}$.

Many interesting combinatorial properties of weakly increasing trees have been investigated recently in [14,15]. One enumerative result proved in [15] is a compact product formula for the number of weakly increasing trees

$$|\mathcal{T}_M| = \frac{1}{1 + N_n} \prod_{i=1}^n \binom{N_i + p_i}{p_i}, \quad (1.2)$$

where $N_i := p_1 + \dots + p_i$ for each $i \in [n]$. A leaf or an internal node (other than the root) of a weakly increasing tree is *old* if it is the rightmost child of its parent; otherwise, it is *young*. Note that old and young nodes on increasing trees were introduced by Chen, Deutsch and Elizalde [9]. As an application of weakly increasing trees, we answer Problem 1.1 completely in the following result.

Theorem 1.3. Fix an integer $p \geq 1$. For any $n \geq 1$,

$$A_n^{(p)}(t) = \sum_{i=0}^{\lfloor \frac{p(n-1)}{2} \rfloor} \gamma_{n,i}^{(p)} t^i (1+t)^{p(n-1)-2i}, \quad (1.3)$$

where $\gamma_{n,i}^{(p)}$ is the number of weakly increasing trees on $\{1^p, 2^p, \dots, (n-1)^p, n\}$ with $i+1$ leaves and without young leaves.

A polynomial $\sum_{i=0}^n a_i t^i$ is *alternatingly increasing* (or has *spiral property*) if

$$a_0 \leq a_n \leq a_1 \leq a_{n-1} \leq \dots \leq a_{\lfloor \frac{n+1}{2} \rfloor}.$$

Obviously, alternatingly increasing is a stronger property than unimodality.

Proposition 1.4 (See [5]). Let $h(t)$ be a polynomial of degree n . Then $h(t)$ can be written uniquely as $h(t) = a(t) + tb(t)$, where $a(t)$ is symmetric with degree n and $b(t)$ is symmetric with degree less than n . More precisely, we have

$$a(t) = \frac{h(t) - t^{n+1}h(1/t)}{1-t} \quad \text{and} \quad b(t) = \frac{t^n h(1/t) - h(t)}{1-t}. \quad (1.4)$$

The decomposition $h(t) = a(t) + tb(t)$ in Proposition 1.4 is called the *symmetric decomposition* of $h(t)$. For example, the symmetric decomposition of $h(t) = 1 + 24t + 120t^2 + 104t^3 + 16t^4$ is

$$h(t) = 1 + 9t + 25t^2 + 9t^3 + t^4 + t(15 + 95t + 95t^2 + 15t^3).$$

As observed in [5], the polynomial $h(t)$ is alternatingly increasing if and only if $a(t)$ and $b(t)$ are both unimodal and have nonnegative coefficients. If $a(t)$ and $b(t)$ are both γ -positive, then we say that $h(t)$ is *bi- γ -positive*. Note that if a polynomial is bi- γ -positive, then it is alternatingly increasing.

Bi- γ -positivity¹ extends γ -positivity from symmetric polynomials to general polynomials. There has been recent interest in studying bi- γ -positivity polynomials arising in combinatorics and geometry; see [2,3,5,12,17]. Not much instances of nonsymmetric bi- γ -positive polynomials in combinatorics were known, these including colored derangement polynomials and binomial Eulerian polynomials for colored permutations (see [3,12]). It would be interesting to find more classes of bi- γ -positivity polynomials that is nonsymmetric. Our next result provides a new class of nonsymmetric bi- γ -positive polynomials.

Theorem 1.5. Fix $p \geq 1$. Let $B_n^{(p)}(t) := A_M(t)$ with $M = \{1^p, 2^p, \dots, n^p, n+1\}$. Then $B_n^{(p)}(t)$ is bi- γ -positive with symmetric decomposition

$$B_n^{(p)}(t) = a_n(t) + tb_n(t), \quad (1.5)$$

where

$$a_n(t) = \left(1 + (pn - p + 1)t + t(1 - t)\frac{\partial}{\partial t}\right)A_n^{(p)}(t) \quad \text{and} \quad b_n(t) = (p - 1)A_n^{(p)}(t).$$

Consequently, $B_n^{(p)}(t)$ is alternatingly increasing.

Remark 1.6. The $p = 2$ case of Theorem 1.5 was proved by Ma, Ma and Yeh [17] using Chen's context-free grammar [8], an approach which seems hard to be adopted for general p . Since context-free grammar deals naturally with labeled structures and if p copies of each positive integer in multipermutations are labeled, then the underlying grammar becomes extremely complicated. We will provide both a computational proof and a short combinatorial proof for the symmetric decomposition (1.5).

Remark 1.7. It should be pointed out that in general the multiset Eulerian polynomial $A_M(t)$ is not bi- γ -positive for any M . Consider for instance $M = \{1^5, 2^2\}$, then

$$A_M(t) = 1 + 10t + 10t^2 = 1 + t + t^2 + t(9 + 9t),$$

which is not bi- γ -positive. However, to determine whether $A_M(t)$ is alternatingly increasing for any M is an interesting open problem.

The rest of this paper is organized as follows. Proofs of Theorems 1.5 and 1.3 are given in Sections 2 and 3, respectively. In Section 4, we prove a recurrence relation for counting weakly increasing trees by internal nodes. Finally, in Section 5, we end this paper by posing an interesting Diophantine equation conjecture arising in this work.

2. Proof of Theorem 1.5

Firstly, we provide both a computational proof and a short combinatorial proof for the symmetric decomposition (1.5).

A Computational Proof of (1.5). Since by (1.4) we have $b_n(t) = \frac{t^{p(n-1)+1}B_n^{(p)}(1/t) - B_n^{(p)}(t)}{1-t}$, the fact that $b_n(t) = (p - 1)A_n^{(p)}(t)$ is equivalent to

$$t^{p(n-1)+1}B_n^{(p)}(1/t) - B_n^{(p)}(t) = (1 - t)(p - 1)A_n^{(p)}(t). \quad (2.1)$$

We aim to verify (2.1) by using MacMahon's identity (1.1).

We need a fundamental result concerning the rational generating functions (see Proposition 4.2.3 and Corollary 4.3.1 in [21]). If $f(n)$ is a polynomial function in n with $\sum_{n \geq 0} f(n)t^n = F(t)$, then

¹ Bi- γ -positivity was called nonsymmetric γ -positivity by Athanasiadis [2, Section 5.1].

$\sum_{n \geq 1} f(-n)t^n = -F(-t^{-1})$. It then follows from (1.1) that

$$\begin{aligned} \frac{-B_n^{(p)}(t)}{(1-t^{-1})^{pn+2}} &= \sum_{k \geq 1} \binom{p-k}{p}^n (-k+1)t^k \\ &= \sum_{k \geq 1} (-1)^{pn+1} \binom{k-1}{p}^n (k-1)t^k \\ &= \sum_{k \geq 0} (-1)^{pn+1} \binom{k+p}{p}^n (k+p)t^{k+p+1}. \end{aligned}$$

Hence,

$$\frac{t^{p(n-1)+1} B_n^{(p)}(t^{-1})}{(1-t)^{pn+2}} = \sum_{k \geq 0} \binom{k+p}{p}^n (k+p)t^k,$$

and so

$$\begin{aligned} &t^{p(n-1)+1} B_n^{(p)}(1/t) - B_n^{(p)}(t) \\ &= (1-t)^{pn+2} \left(\sum_{k \geq 0} \binom{k+p}{p}^n (k+p)t^k - \sum_{k \geq 0} \binom{k+p}{p}^n (k+1)t^k \right) \\ &= (1-t)^{pn+2} (p-1) \sum_{k \geq 0} \binom{k+p}{p}^n t^k \\ &= (1-t)(p-1) A_n^{(p)}(t), \end{aligned}$$

which proves (2.1). Thus, $b_n(t) = (p-1)A_n^{(p)}(t)$.

It remains to prove that $a_n(t) = \left(1 + (pn-p+1)t + t(1-t)\frac{\partial}{\partial t}\right) A_n^{(p)}(t)$. By (1.1), we have

$$\frac{\left(1 + pnt + t(1-t)\frac{\partial}{\partial t}\right) A_n^{(p)}(t)}{(1-t)^{pn+2}} = \sum_{k \geq 0} \binom{p+k}{k}^n (k+1)t^k = \frac{B_n^{(p)}(t)}{(1-t)^{pn+2}},$$

which implies

$$B_n^{(p)}(t) = \left(1 + pnt + t(1-t)\frac{\partial}{\partial t}\right) A_n^{(p)}(t).$$

This completes the proof of (1.5). $\square$

The above proof of (1.5) is computational, below we provide a short combinatorial one.

A Combinatorial Proof of (1.5). Let $B_{n,k}^{(p)} = [t^k]B_n^{(p)}(t)$ and $A_{n,k}^{(p)} = [t^k]A_n^{(p)}(t)$. Then, $B_{n,k}^{(p)}$ (resp. $A_{n,k}^{(p)}$) counts the number of permutations of $\{1^p, \dots, n^p, n+1\}$ (resp. $\{1^p, \dots, n^p\}$) with k descents. Since every permutation counted by $B_{n,k}^{(p)}$ can be constructed

- either from a permutation counted by $A_{n,k}^{(p)}$ by inserting $n+1$ into one of its k descent positions or the end;
- or from a permutation counted by $A_{n,k-1}^{(p)}$ by inserting $n+1$ into one of its $pn-k$ non-descent positions or the front,

we have

$$B_{n,k}^{(p)} = (k+1)A_{n,k}^{(p)} + (pn+1-k)A_{n,k-1}^{(p)}.$$

The above recurrence relation can be rewritten as $B_{n,k}^{(p)} = a_{n,k} + b_{n,k-1}$ where

$$a_{n,k} := (k+1)A_{n,k}^{(p)} + (p(n-1)+2-k)A_{n,k-1}^{(p)} \quad \text{and} \quad b_{n,k-1} := (p-1)A_{n,k-1}^{(p)}.$$

Since $A_{n,k}^{(p)} = A_{n,p(n-1)-k}^{(p)}$ (i.e., $A_n^{(p)}(t)$ is symmetric of degree $p(n-1)$ as proved in [6]), we have

$$a_{n,p(n-1)+1-k} = (p(n-1)+2-k)A_{n,p(n-1)+1-k}^{(p)} + (k+1)A_{n,p(n-1)-k}^{(p)} = a_{n,k},$$

which means that the polynomial $\sum_{k=0}^{p(n-1)+1} a_{n,k} t^k$ is symmetric of degree $p(n-1)+1$. This proves that (1.5) is the symmetric decomposition of $B_n^{(p)}(t)$. $\square$

In order to prove the bi- γ -positivity of $B_n^{(p)}(t)$, we need a criterion for the real-rootedness of the linear extension of two interacting real polynomials. Let f and g be two real-rooted polynomials with $\deg f = \deg g + 1 = n$. If $\{r_i\}$ and $\{s_j\}$ are all zeros of f and g in nonincreasing order, then we say that g interlaces f if

$$r_n \leq s_{n-1} \leq \dots \leq s_2 \leq r_2 \leq s_1 \leq r_1.$$

Lemma 2.1 (Wang and Yeh [22]). Suppose that $f(t)$ and $g(t)$ are two real-rooted polynomials with nonnegative coefficients and g interlaces f . Let

$$F(t) = (at + b)f(t) + t(ct + d)g(t).$$

If $ad \geq bc$, then $F(t)$ has also only real zeros.

Now we are ready for the proof of Theorem 1.5.

Proof of Theorem 1.5. We have already proved that (1.5) is the symmetric decomposition of $B_n^{(p)}(t)$. Thus, it is sufficient to show that $a_n(t)$ and $b_n(t)$ both have only real zeros by Brändén's observation [4, Remark 7.3.1]. By a result of Simion [19], it is clear that $b_n(t) = (p-1)A_n^{(p)}(t)$ is real-rooted. Recall that

$$a_n(t) = \left(1 + (pn - p + 1)t + t(1 - t) \frac{\partial}{\partial t} \right) A_n^{(p)}(t). \quad (2.2)$$

Note that $h'(t)$ interlaces $h(t)$ for any real-rooted polynomial $h(t)$. Since (2.2) satisfies the criterion of Lemma 2.1, $a_n(t)$ has only real roots. This completes the proof. $\square$

3. Proof of Theorem 1.3

This section is devoted to the proof of Theorem 1.3. For a tree $T \in \mathcal{T}_M$, denote by $\text{int}(T)$ the number of internal nodes in T , where an *internal node* of T is a node that has at least one child. The following result was proved in [15] by introducing a generalization of the Foata–Strehl group action [11] on weakly increasing trees.

Theorem 3.1 (Lin–Ma–Ma–Zhou [15]). For a fixed multiset $M = \{1^{p_1}, 2^{p_2}, \dots, n^{p_n}\}$ with $p = p_1 + \dots + p_n$, the M -Eulerian–Narayana polynomial has γ -positivity expansion

$$\sum_{T \in \mathcal{T}_M} t^{\text{int}(T)-1} = \sum_{i=0}^{\lfloor \frac{p-1}{2} \rfloor} \gamma_{M,i} t^i (1+t)^{p-1-2i},$$

where $\gamma_{M,i}$ enumerates weakly increasing trees on M with $i+1$ leaves and without young leaves.

A weakly increasing tree is called a *multiset tree* if no child has the same label as its parent. Let $\mathcal{P}_M$ be the set of all multiset trees in $\mathcal{T}_M$. The following result justifies the name multiset trees.

Theorem 3.2 (Lin–Ma–Ma–Zhou [15]). There exists a natural bijection $\Psi : \mathfrak{S}_M \rightarrow \mathcal{P}_M$ such that $\text{asc}(\pi) = \text{int}(\Psi(\pi)) - 1$ for each $\pi \in \mathfrak{S}_M$.

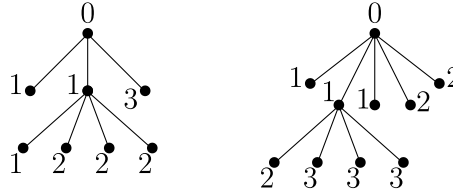


Fig. 2. (Left) A weakly increasing tree in $\mathcal{T}_3^{(3)}$ and (Right) a multiset tree in $\mathcal{P}_3^{(3)}$ with 2 internal nodes and type $((1, 1, 0), (0, 0, 1))$.

The bijection $\Psi : \mathfrak{S}_M \rightarrow \mathcal{P}_M$ is a generalization of the classical encoding of permutations as increasing trees in [21, p. 25]. We provide the brief description of Ψ for the reader's convenience. For any $\pi \in \mathfrak{S}_M$, construct the tree $\Psi(\pi)$ on $M \cup \{0\}$ by defining node i to be the child of the rightmost element j , $j < i$, of π which precedes i . If there is no such element j , then let i be the child of the root 0.

Let $\mathcal{P}_n^{(p)}$ be the set of multiset trees on $\{1^p, 2^p, \dots, n^p\}$. In view of Theorem 3.2, we have the following new interpretation of the p -multiset Eulerian polynomials

$$A_n^{(p)}(t) = \sum_{T \in \mathcal{P}_n^{(p)}} t^{\text{int}(T)-1}. \quad (3.1)$$

Let $\mathcal{T}_n^{(p)}$ be the set of weakly increasing trees on $\{1^p, 2^p, \dots, (n-1)^p, n\}$. The following equidistribution is our main result in this section.

Theorem 3.3. For integers $p, n \geq 1$,

$$\sum_{T \in \mathcal{P}_n^{(p)}} t^{\text{int}(T)} = \sum_{T \in \mathcal{T}_n^{(p)}} t^{\text{int}(T)}. \quad (3.2)$$

Combining (3.2) and (3.1) gives

$$A_n^{(p)}(t) = \sum_{T \in \mathcal{T}_n^{(p)}} t^{\text{int}(T)-1},$$

which implies Theorem 1.3 in view of Theorem 3.1. In order to prove Theorem 3.3, we need to introduce a refinement of it by degree types of trees.

3.1. Degree types of trees and a refinement of Theorem 3.3

For a tree $T \in \mathcal{T}_M$ with $M = \{1^{p_1}, \dots, n^{p_n}\}$ and a letter $k \in [n]$, we say that k has degree type $\mathbf{t}_k = (t_1, t_2, \dots, t_{p_k})$ if there are t_i ($1 \leq i \leq p_k$) nodes that have exactly i children labeled by k in T . For example, the letter 2 in the tree in Fig. 1 has degree type $(2, 1, 0, 0)$. Note that if T is a multiset tree, then letter 1 always has degree type $(0, \dots, 0, 1)$, as all nodes labeled by 1 must be the children of the root. Thus, we say a multiset tree $T \in \mathcal{P}_n^{(p)}$ has (letter-degree) type $\mathbf{t} = (\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_{n-1})$ if the letter $i+1$ has degree type $\mathbf{t}_i$ in T for $1 \leq i \leq n-1$. On the other hand, for a tree $T \in \mathcal{T}_n^{(p)}$, we say T has (letter-degree) type $\mathbf{t} = (\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_{n-1})$ if the letter i has degree type $\mathbf{t}_i$ in T for $1 \leq i \leq n-1$. The following result is a refinement of Theorem 3.3 by degree types.

Theorem 3.4 (Refined Theorem 3.3 by Degree Types). For any integer k and type $\mathbf{t}$, there are equal number of multiset trees in $\mathcal{P}_n^{(p)}$ and weakly increasing trees in $\mathcal{T}_n^{(p)}$ with k internal nodes and type $\mathbf{t}$.

For example, there are 12 trees in $\mathcal{T}_3^{(3)}$ (resp. $\mathcal{P}_3^{(3)}$) with 2 internal nodes and type $((1, 1, 0), (0, 0, 1))$, one of which is drawn in left (resp. right) side of Fig. 2.

Denote by $\mathcal{G}_{\mathbf{t},k}$ (resp. $\mathcal{H}_{\mathbf{t},k}$) the set of trees in $\mathcal{P}_n^{(p)}$ (resp. $\mathcal{T}_n^{(p)}$) with k internal nodes and type $\mathbf{t}$. Let $G_{\mathbf{t},k} = |\mathcal{G}_{\mathbf{t},k}|$ and $H_{\mathbf{t},k} = |\mathcal{H}_{\mathbf{t},k}|$. In the rest of this section, we aim to verify that $G_{\mathbf{t},k}$ and $H_{\mathbf{t},k}$ share the same recurrence relation. Throughout this section, a rooted tree of height one will be called a *small tree*. The size of a small tree is the number of its edges.

3.2. The degree type of multiset trees: a recurrence relation for $G_{\mathbf{t},k}$

Lemma 3.5. Let $\mathbf{t} = (\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_{n-1})$ where $\mathbf{t}_i \in \mathbb{N}^p$. Let $\mathbf{t}' = (\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_{n-1}, \mathbf{t}_n)$ where $\mathbf{t}_n = (n_1, n_2, \dots, n_p) \in \mathbb{N}^p$ satisfying $n_1 + n_2 + \dots + n_p = m$ and $n_1 + 2n_2 + \dots + pn_p = p$. Then

$$G_{\mathbf{t}',k} = \sum_{l \leq k} G_{\mathbf{t},l} \binom{pn+1-l}{k-l} \binom{l}{m-k+l} \binom{m}{n_1, n_2, \dots, n_p}. \quad (3.3)$$

Proof. Since multiset trees have the property that any node cannot be the parent of another node with the same label, for any tree in $\mathcal{G}_{\mathbf{t}',k}$ there are exactly n_i small trees of size i ($1 \leq i \leq p$) induced by all the nodes labeled by $n+1$ and their parents. Thus, any multiset tree in $\mathcal{G}_{\mathbf{t}',k}$ can be constructed from a multiset tree in $\mathcal{G}_{\mathbf{t},l}$ ($l \leq k$) by

- first choosing $k-l$ leaves and $m-k+l$ internal nodes,
- and then attaching m small trees (among which are n_i small trees of size i) to these m chosen nodes.

Since there are $\binom{pn+1-l}{k-l}$ ways to choose $k-l$ leaves, $\binom{l}{m-k+l}$ ways to choose $m-k+l$ internal nodes and $\binom{m}{n_1, n_2, \dots, n_p}$ ways to attach these m small trees, the desired recurrence relation follows. $\square$

3.3. A lemma for counting plane trees with given degree type

We need a lemma to count some kind of plane trees with a given degree type before we can deduce a recurrence relation for $H_{\mathbf{t},k}$. A plane tree with n_i nodes that each has exactly i children for $i = 1, 2, \dots$ is said to have *degree type* $(n_1, n_2, \dots)$. A plane tree is *proper* if no child of its root is a leaf. We aim to count proper plane trees with given degree type using a general bijective algorithm for trees due to Chen [7].

Lemma 3.6. The number of proper plane trees on $n+i+1$ nodes with root degree i and degree type $(n_1, n_2, \dots, n_{i-1}, n_i+1, n_{i+1}, \dots)$ is

$$\frac{i(m-1)!}{n_1!n_2! \dots} \binom{n}{m-i},$$

where $n = \sum_{i \geq 1} in_i$ and $m = \sum_{i \geq 1} n_i$.

Following Chen [7], a *Schröder tree* is a labeled rooted tree for which the set of subtrees of any vertex is endowed with the structure of ordered partitions. His bijection between forests of k small trees on $[n+k-1]$ and Schröder trees on $[n]$ with k blocks can be described in the following steps. Given such a forest F of k small trees:

- (1) Mark all nodes $n+1, n+2, \dots, n+k-1$ in F by the symbol $*$.
- (2) Find the tree T in F with no marked node and has the smallest root. Let i be the root of T .
- (3) Find the tree T^* in F that contains the smallest marked node. Let j^* be this marked node.
- (4) There are two ways to merge T and T^* called *horizontal merge* and *vertical merge*.

- (i) **Horizontal merge:** If j^* is the root of T^* , then merge T and T^* by identifying i and j^* , keep i as the new node, and put the subtree T^* at the right-hand side of T .

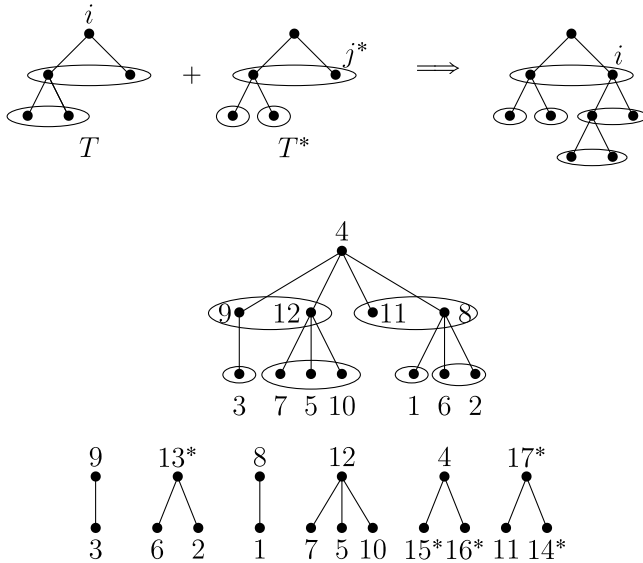
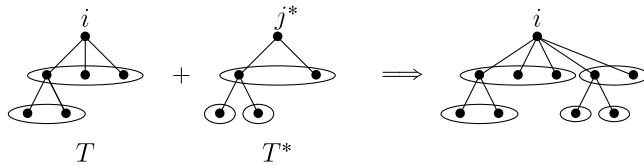


Fig. 3. A Schröder tree on $\{1, 2, \dots, 12\}$ and its small tree decomposition.



(ii) **Vertical merge:** If j^* is the leaf of T^* , then merge T and T^* by identifying i and j^* and keep i as the new node.

(5) Repeat the above procedure until F becomes a Schröder tree.

For an example of the above construction, see Fig. 3. For the sake of convenience, the nodes that are marked by $*$ in step (1) above are called *marked nodes*. Using Chen's bijection, we can now prove Lemma 3.6.

Proof of Lemma 3.6. We will label the proper plane trees in concern by $[n + i + 1]$, count them using Chen's bijection and divide the resulting number by $(n + i + 1)!$.

If a node v of a Schröder tree is not a leaf, then all children of v are called the *fiber* of v . A labeled plane tree can be viewed as a Schröder tree such that every fiber having exactly one block. In Chen's construction of a Schröder tree T from a forest of small trees, there is a fiber of T with more than one block if and only if a horizontal merge occurs, namely, a small tree in the forest has marked root. Thus, under Chen's bijection, labeled proper plane trees with root degree i and degree type $(n_1, n_2, \dots, n_{i-1}, n_i + 1, n_{i+1}, \dots)$ are in one-to-one correspondence with forests of $m + 1$ small trees such that

- all roots of small trees are unmarked,
- the small tree with greatest marked leaf must have size i and all the other leaves are marked (the root of this small tree becomes the root of the plane tree and the require of 'proper' forces all the other leaves must be marked),
- there are exactly $n_i + 1$ small trees of size i and n_j small trees of size j for $j \neq i$.

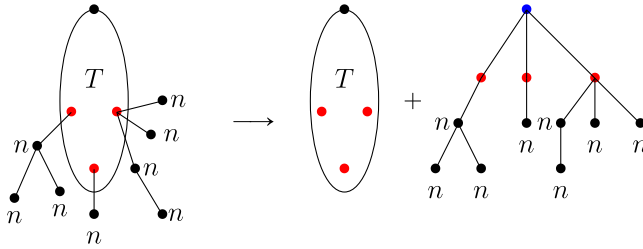


Fig. 4. The pair decomposition of weakly increasing trees.

Since all roots of the $m + 1$ small trees are unmarked, they must be chosen from $[n + i + 1]$ and so there are $\binom{n+i+1}{n_0, n_1, n_2, \dots, n_i+1, \dots}$ ways to choose all the roots, where $n_0 := n + i + 1 - (m + 1) = n + i - m$. Once the roots are chosen, there are $(n_i + 1) \binom{m-1}{i-1} i!$ ways to form the small tree with greatest marked leaf and $n!$ ways to form all the other m small trees. Therefore, the number of labeled proper plane trees in concern is

$$(n_i + 1) \binom{n+i+1}{n_0, n_1, n_2, \dots, n_i+1, \dots} \binom{m-1}{i-1} n! i!.$$

Dividing by $(n + i + 1)!$ yields the desired number for unlabeled proper plane trees. $\square$

3.4. The degree type of weakly increasing trees: a recurrence relation for $H_{t,k}$

In this section, a weakly increasing tree T on $\{1^{p_1}, 2^{p_2}, \dots, n^{p_n}\}$ is said to have (letter-degree) type $\mathbf{t} = (t_1, t_2, \dots, t_n)$ if the letter i has degree type t_i in T for $1 \leq i \leq n$. Denote by $\mathcal{A}_{t,k}$ the set of weakly increasing trees with k internal nodes and type $\mathbf{t}$. Let $A_{t,k}$ be the cardinality of $\mathcal{A}_{t,k}$. We have the following recurrence relation for $A_{t,k}$.

Lemma 3.7. Let $\mathbf{t} = (t_1, t_2, \dots, t_{n-1})$ where $t_i \in \mathbb{N}^p$. Let $\mathbf{t}' = (t_1, t_2, \dots, t_{n-1}, t_n)$ where $t_n = (n_1, n_2, \dots, n_p) \in \mathbb{N}^p$ satisfying $n_1 + n_2 + \dots + n_p = m$ and $n_1 + 2n_2 + \dots + pn_p = p$. Then

$$A_{t',k} = \sum_{l \leq k} A_{t,l} \binom{m}{n_1, n_2, \dots, n_p} \binom{pn+1-l}{k-l} \binom{l}{l+m-k} \left(1 - \frac{p(k-l)}{m(pn+1-l)}\right). \quad (3.4)$$

Proof. Every weakly increasing tree $T' \in \mathcal{A}_{t',k}$ can be decomposed uniquely into a pair (T, P) , where

- T is the subtree of T' induced by the nodes with labels less than n and the nodes that have at least a child n in T' are marked in red;
- and P is the plane tree formed by attaching all the subtrees of T' induced by the nodes labeled n and their parents to a (new) root.

See Fig. 4 for the illustration of this decomposition, which is called the *pair decomposition* of weakly increasing trees. Notice that P is a proper plane trees on $p + i + 1$ nodes with root degree i for some $1 \leq i \leq m$ (in fact i is the number of marked nodes in T) and degree type $(n_1, n_2, \dots, n_{i-1}, n_i + 1, n_{i+1}, \dots, n_p)$, while T is a weakly increasing tree in $\mathcal{A}_{t,l}$ for some $l \leq k$ with i marked nodes, among which $l + m - k$ ones are internal nodes of T and $k + i - l - m$ ones are leaves of T .

Since the pair decomposition is reversible, it then follows from Lemma 3.6 that $A_{t',k}$ satisfies the recurrence relation

$$A_{t',k} = \sum_{l \leq k} A_{t,l} \sum_{i=1}^m \frac{i(m-1)!}{n_1! n_2! \dots n_p!} \binom{p}{m-i} \binom{l}{l+m-k} \binom{(n-1)p+1-l}{k+i-l-m}.$$

Applying the simple combinatorial identity proved in [Lemma 3.8](#) then gives the desired recurrence relation for $A_{t',k}$. $\square$

Lemma 3.8. For any integers $k \geq l \geq 0$ and $m, n, p \geq 1$, we have

$$\sum_i i \binom{p}{m-i} \binom{(n-1)p+1-l}{k+i-l-m} = m \binom{np+1-l}{k-l} - p \binom{np-l}{k-l-1}. \quad (3.5)$$

Proof. Using the Vandermonde identity

$$\sum_k \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r},$$

we have

$$\begin{aligned} & \sum_i i \binom{p}{m-i} \binom{(n-1)p+1-l}{k+i-l-m} \\ &= \sum_i (m - (m-i)) \binom{p}{m-i} \binom{(n-1)p+1-l}{k+i-l-m} \\ &= \sum_i m \binom{p}{m-i} \binom{(n-1)p+1-l}{k+i-l-m} - \sum_j j \binom{p}{j} \binom{(n-1)p+1-l}{k-l-j} \\ &= m \binom{np+1-l}{k-l} - \sum_j p \binom{p-1}{j-1} \binom{(n-1)p+1-l}{k-l-j} \\ &= m \binom{np+1-l}{k-l} - p \binom{np-l}{k-l-1}, \end{aligned}$$

as desired. $\square$

Now we are ready to verify that $H_{t,k}$ share the same recurrence relation as $G_{t,k}$ in [Lemma 3.5](#), which proves [Theorem 3.4](#).

Lemma 3.9. Let $\mathbf{t} = (\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_{n-1})$ where $\mathbf{t}_i \in \mathbb{N}^p$. Let $\mathbf{t}' = (\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_{n-1}, \mathbf{t}_n)$ where $\mathbf{t}_n = (n_1, n_2, \dots, n_p) \in \mathbb{N}^p$ satisfying $n_1 + n_2 + \dots + n_p = m$ and $n_1 + 2n_2 + \dots + pn_p = p$. Then

$$H_{t',k} = \sum_{l \leq k} H_{t,l} \binom{pn+1-l}{k-l} \binom{l}{m-k+l} \binom{m}{n_1, n_2, \dots, n_p}. \quad (3.6)$$

Proof. Since every weakly increasing tree on $\{1^p, 2^p, \dots, n^p, n+1\}$ can be constructed from a unique weakly increasing tree on $\{1^p, 2^p, \dots, n^p\}$ by attaching the node $n+1$, we have

$$H_{t',k} = kA_{t',k} + (pn+2-k)A_{t',k-1}. \quad (3.7)$$

Similarly, we also have

$$H_{t,l} = lA_{t,l} + (p(n-1)+2-l)A_{t,l-1}. \quad (3.8)$$

On the one hand, it follows from (3.8) that the right-hand side of (3.6) dividing $\binom{m}{n_1, \dots, n_p}$ equals

$$\begin{aligned} & \sum_{l \leq k} (lA_{t,l} + (p(n-1)+2-l)A_{t,l-1}) \binom{pn+1-l}{k-l} \binom{l}{m-k+l} \\ &= \sum_{l \leq k} lA_{t,l} \binom{pn+1-l}{k-l} \binom{l}{m-k+l} + \sum_{l \leq k} (p(n-1)+1-l)A_{t,l} \binom{pn-l}{k-l-1} \binom{l+1}{m+1-k+l}, \end{aligned}$$

which can be simplified to

$$\sum_{l \leq k} A_{t,l} \binom{pn+1-l}{k-l} \binom{l}{m-k+l} \left(l + \frac{(p(n-1)+1-l)(k-l)(l+1)}{(pn+1-l)(m+1+l-k)} \right). \quad (3.9)$$

On the other hand, it follows from (3.7) and (3.4) that $H_{t',k}/\binom{m}{n_1, \dots, n_p}$ equals

$$\begin{aligned} & (kA_{t',k} + (pn+2-k)A_{t',k-1}) / \binom{m}{n_1, \dots, n_p} \\ &= \sum_{l \leq k} kA_{t,l} \binom{pn+1-l}{k-l} \binom{l}{l+m-k} \left(1 - \frac{p(k-l)}{m(pn+1-l)} \right) \\ &+ \sum_{l \leq k-1} (pn+2-k)A_{t,l} \binom{pn+1-l}{k-l-1} \binom{l}{l+m+1-k} \left(1 - \frac{p(k-l-1)}{m(pn+1-l)} \right), \end{aligned}$$

which is again simplified to (3.9). This completes the proof of the lemma. $\square$

4. A recurrence relation for the M -Eulerian–Narayana numbers

For a multiset M , define the M -Eulerian–Narayana number $A_{M,k}$ as the number of weakly increasing trees in $\mathcal{T}_M$ with k internal nodes. In Lemma 3.7, we have found a recurrence relation for the letter-type refinement of the M -Eulerian–Narayana numbers $A_{t,k}$ by using the pair decomposition of weakly increasing trees (see Fig. 4). In this section, we modify this decomposition to the triple decomposition of weakly increasing trees (see Fig. 6) and derive the following recurrence relation for the M -Eulerian–Narayana numbers $A_{M,k}$.

Theorem 4.1. Let $M = \{1^{p_1}, 2^{p_2}, \dots, (n-1)^{p_{n-1}}\}$ and $M' = M \cup \{n^{p_n}\}$. Let $p = p_1 + p_2 + \dots + p_{n-1}$ and $p' = p + p_n$. Then

$$A_{M',k} = \sum_{l \leq k} A_{M,l} \binom{p_n+l-1}{k-1} \binom{p'+1-l}{k-l} \left(1 - \frac{(p_n+l)(k-l)}{k(p'+1-l)} \right) \quad (4.1)$$

with initial condition $A_{\emptyset,l} = \delta_{l,0}$.

Remark 4.2. It follows from (4.1) that

$$A_{\{1^n\},k} = \frac{1}{n+1} \binom{n-1}{k-1} \binom{n+1}{k}, \quad (4.2)$$

the Narayana numbers and

$$A_{[n],k} = kA_{[n-1],k} + (n+1-k)A_{[n-1],k-1},$$

the classical two term recurrence relation for the Eulerian numbers. Note that the symmetry

$$A_{M',k} = A_{M',p'+1-k}$$

is also evident from (4.1). The M -Eulerian–Narayana polynomial $\sum_{k \geq 1} A_{M,k} t^k$ was proved in [15] to be γ -positive and further conjectured to be real-rooted for any M . It was our attempt to derive a two term recurrence relation for $A_{M,k}$ to prove this real root conjecture that leads to (4.1).

An interesting application of Theorem 4.1 is the following equidistribution.

Corollary 4.3. For any $n \geq 0$, we have the equidistribution

$$\sum_{T \in \mathcal{T}_{\{1^{2n+2}, 2^n\}}} t^{\text{int}(T)} = \sum_{T \in \mathcal{T}_{\{1^{2n+1}, 2^{n+1}\}}} t^{\text{int}(T)}.$$

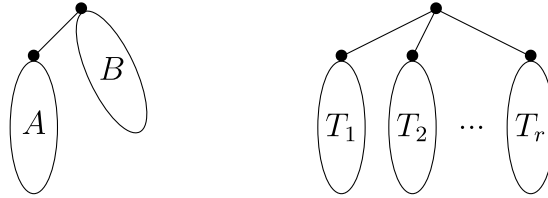


Fig. 5. Two decompositions of plane trees.

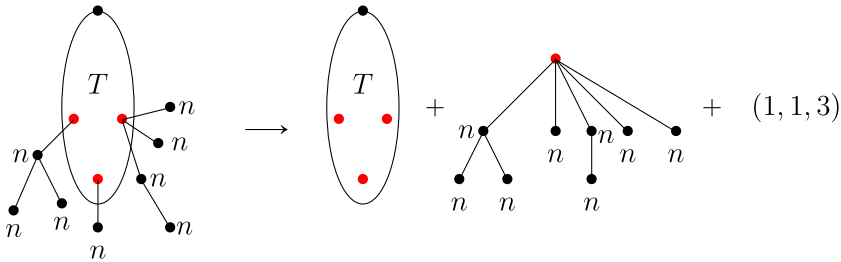


Fig. 6. The triple decomposition of weakly increasing trees.

Proof. Using (4.1) and involving (4.2), the above equidistribution is equivalent to the following combinatorial identity (for $n \geq 0$ and $1 \leq k \leq 3n + 2$):

$$\begin{aligned} \sum_{l=1}^k \binom{n+l-1}{k-1} \binom{3n+3-l}{k-l} \left(1 - \frac{(n+l)(k-l)}{k(3n+3-l)}\right) \frac{\binom{2n+1}{l-1} \binom{2n+3}{l}}{2n+3} \\ = \sum_{l=1}^k \binom{n+l}{k-1} \binom{3n+3-l}{k-l} \left(1 - \frac{(n+1+l)(k-l)}{k(3n+3-l)}\right) \frac{\binom{2n}{l-1} \binom{2n+2}{l}}{2n+2}, \end{aligned}$$

which can be verified by Maple. $\square$

In order to prove Theorem 4.1, we need some preparations. Let $\mathcal{T}_{n,r}$ denote the set of plane trees with n edges and root degree r . Introduce the polynomial $T_{n,r}(t)$ as

$$T_{n,r}(t) = \sum_{T \in \mathcal{T}_{n,r}} t^{\text{int}(T)-1}.$$

Define the generating function

$$N = N(t, x) := \sum_{n \geq 0} x^{n+1} \sum_{T \in \mathcal{T}_n} t^{\text{int}(T)} = x + tx^2 + (t + t^2)x^3 + (t + 3t^2 + t^3)x^4 + \cdots,$$

where $\mathcal{T}_n$ is the set of plane trees with n edges. By the first decomposition of plane trees in Fig. 5 we have

$$N = x + xtN + N(N - x).$$

Thus, the compositional inverse of N is

$$N^{(-1)} = \frac{x - x^2}{1 + tx - x}.$$

By the second decomposition of plane trees with root degree r in Fig. 5 and the Lagrange inversion formula (see [20, Theorem 5.4.2]) we have (for $r \geq 1$)

$$\begin{aligned} T_{n,r}(t) &= [x^n] N^r \\ &= \frac{r}{n} [x^{n-r}] \left(\frac{1+x(t-1)}{1-x} \right)^n \\ &= \frac{r}{n} \sum_k (-1)^k \binom{-n}{k} \binom{n}{n-r-k} (t-1)^{n-r-k} \\ &= \frac{r}{n} \sum_k \binom{n+k-1}{k} \binom{n}{n-r-k} (t-1)^{n-r-k}. \end{aligned}$$

Therefore, we have proved the following expression for $T_{n,r}(t)$.

Lemma 4.4. For $n, r \geq 1$, we have

$$T_{n,r}(t) = \frac{r}{n} \sum_k \binom{n+k-1}{k} \binom{n}{r+k} (t-1)^{n-r-k}. \quad (4.3)$$

We also need the following combinatorial identities.

Lemma 4.5. For any integers $k, j \geq 0$, we have

$$\sum_r r x^{r-j} \binom{k-1+r}{k-1+j} = \frac{x(k+j) + j(1-x)}{(1-x)^{k+j+1}}. \quad (4.4)$$

Proof. The above equation is equivalent to

$$\sum_r (r+j)x^r \binom{k-1+r+j}{k-1+j} = \frac{x(k+j)}{(1-x)^{k+j+1}} + \frac{j}{(1-x)^{k+j}},$$

which is an easy consequence of the binomial theorem $(1-x)^{-k-1} = \sum_i \binom{k+i}{i} x^i$. $\square$

Lemma 4.6. For any integers $l \geq 1$ and $j, s \geq 0$, we have

$$\sum_r \binom{l+r-1}{l+j-1} \binom{n-1+s-r}{n-1} r = \frac{(l+j)s + jn}{s-j} \binom{n+s+l-1}{n+l+j}. \quad (4.5)$$

Proof. By the binomial theorem, the left-hand side of (4.5) equals

$$\begin{aligned} & [x^{n-1}] \left((1+x)^{n-1+s-j} \sum_r r \binom{l+r-1}{l+j-1} \frac{1}{(1+x)^{r-j}} \right) \\ &= [x^{n-1}] \left((1+x)^{n-1+s-j} \frac{\frac{l+j}{1+x} + \frac{jx}{1+x}}{\left(1 - \frac{1}{1+x}\right)^{l+j+1}} \right) \quad (\text{by Lemma 4.5}) \\ &= [x^{n+l+j}] (1+x)^{n+s+l-1} (l+j+jx) \\ &= (l+j) \binom{n+s+l-1}{n+l+j} + j \binom{n+s+l-1}{n+l+j-1}, \end{aligned}$$

which is the right-hand side of (4.5). $\square$

Now we are ready for the proof of Theorem 4.1.

Proof of Theorem 4.1. Every weakly increasing tree $T' \in \mathcal{T}_M'$ can be decomposed uniquely into a triple (T, P, h) , where

- T is the subtree of T' induced by the nodes with labels less than n and the nodes that have at least a child n in T' are marked in red;
- and $P \in \mathcal{T}_{p_n}$ is the plane tree formed by attaching all the subtrees of T' induced by the nodes labeled n to a (new) root;
- if P has r branches at the root and T has m marked nodes, then $h = (h_1, \dots, h_m)$ is the m -composition of r , where h_i is the number of the children labeled n of the i th (in a specified order, e.g., in depth-first order) marked node in T' .

See Fig. 6 for the illustration of this decomposition, which is a variation of the pair decomposition in Fig. 4 and will be called the *triple decomposition* of weakly increasing trees. The triple decomposition is reversible and satisfies

$$\text{int}(T') = \text{int}(T) + (\text{int}(P) - 1) + \#\{\text{marked leaves of } T\}.$$

Conversely, for any weakly increasing tree $T \in \mathcal{T}_M$ with l internal nodes, we can

- choose i marked nodes from these l internal nodes of T in $\binom{l}{i}$ ways and choose $m - i$ marked nodes from the remaining $p + 1 - l$ leaves of T in $\binom{p+1-l}{m-i}$ ways;
- from a fixed plane tree $P \in \mathcal{T}_{p_n, r}$, there are $\binom{r-1}{m-1}$ ways to partition the r branches of P into m ordered parts (i.e., to choose a m -composition of r in $\binom{r-1}{m-1}$ ways) and then attach these parts to the chosen marked nodes.

Thus, we write

$$t^l \rightsquigarrow \sum_{m, i, r} \binom{l}{i} \binom{p+1-l}{m-i} \binom{r-1}{m-1} T_{p_n, r}(t) t^{l+m-i}, \quad (4.6)$$

which means that from any tree T in $\mathcal{T}_M$ with weight t^l (i.e., with l internal nodes), the enumerator of the trees in $\mathcal{T}_{M'}$ generated from T by the weight $t^{\#\text{internal nodes}}$ is exactly the summation in (4.6). It remains to show that this summation can be simplified to

$$\sum_i t^{l+i} \binom{p_n+l-1}{l+i-1} \binom{p'+1-l}{i} \left(1 - \frac{i(p_n+l)}{(l+i)(p'+1-l)}\right), \quad (4.7)$$

from which (4.1) follows in view of the generating rule (4.6).

By the Vandermonde identity we have

$$\begin{aligned} & \sum_{m, i, r} \binom{l}{i} \binom{p+1-l}{m-i} \binom{r-1}{m-1} T_{p_n, r}(t) t^{l+m-i} \quad (\text{set } m-i=j) \\ &= \sum_{j, r} t^{l+j} \binom{p+1-l}{j} T_{p_n, r}(t) \sum_i \binom{l}{l-i} \binom{r-1}{i+j-1} \\ &= \sum_{j, r} t^{l+j} \binom{p+1-l}{j} T_{p_n, r}(t) \binom{l+r-1}{l+j-1}. \end{aligned}$$

Using the expression (4.3) for $T_{n, r}(t)$, the above summation becomes

$$\begin{aligned} & \sum_{j, k, r} t^{l+j} \binom{p+1-l}{j} \binom{l+r-1}{l+j-1} \frac{r}{p_n} \binom{p_n+k-1}{k} \binom{p_n}{r+k} (t-1)^{p_n-r-k} \quad (\text{set } r+k=s) \\ &= \sum_{j, s} t^{l+j} \binom{p+1-l}{j} (t-1)^{p_n-s} \frac{1}{p_n} \binom{p_n}{s} \sum_r \binom{l+r-1}{l+j-1} \binom{p_n-1+s-r}{p_n-1} r \\ &= \sum_{j, s} t^{l+j} \binom{p+1-l}{j} (t-1)^{p_n-s} \frac{1}{p_n} \binom{p_n}{s} \frac{(l+j)s + jp_n}{s-j} \binom{p_n+s+l-1}{p_n+l+j} \quad (\text{by Lemma 4.6}) \end{aligned}$$

$$\begin{aligned}
 &= \sum_{i,j,s} (-1)^{p_n-s-i} t^{l+i+j} \frac{1}{p_n} \binom{p_n}{s} \binom{p_n-s}{i} \binom{p+1-l}{j} \frac{(l+j)s+jp_n}{s-j} \binom{p_n+s+l-1}{p_n+l+j} \\
 &= \sum_{i,j} t^{l+i+j} \frac{1}{p_n} \binom{p_n}{i} \binom{p+1-l}{j} \sum_s (-1)^{p_n-s-i} \frac{(l+j)s+jp_n}{s-j} \binom{p_n-i}{s} \binom{p_n+s+l-1}{p_n+l+j},
 \end{aligned}$$

which can be further simplified to

$$\sum_{i,j} t^{l+i+j} \left(1 - \frac{i(p_n+l)}{p_n(l+i+j)}\right) \binom{p_n}{i} \binom{p+1-l}{j} \binom{p_n+l-1}{l+i+j-1}, \quad (4.8)$$

as

$$\begin{aligned}
 &\sum_s (-1)^{p_n-s-i} \binom{p_n-i}{s} \binom{p_n+s+l-1}{p_n+l+j} \left(\frac{(l+j)s+jp_n}{s-j}\right) \\
 &= [x^{p_n+l+j-1}] \left(\frac{(1+x)^{p_n+l-1}}{p_n+l+j} \sum_s (-1)^{p_n-s-i} \binom{p_n-i}{s} (1+x)^s ((l+j)s+jp_n) \right) \\
 &= [x^{l+i+j}] \left(\frac{(1+x)^{p_n+l-1}}{p_n+l+j} (jp_n x + (l+j)(p_n-i)(1+x)) \right) \\
 &= \frac{(l+j)p_n - li}{l+i+j} \binom{p_n+l-1}{l+i+j-1}.
 \end{aligned}$$

After setting $i+j=k$, summation (4.8) becomes

$$\sum_k t^{l+k} \binom{p_n+l-1}{l+k-1} \sum_i \left(1 - \frac{i(p_n+l)}{p_n(l+k)}\right) \binom{p_n}{i} \binom{p+1-l}{k-i},$$

which equals (4.7) in view of

$$\begin{aligned}
 &\sum_i \left(1 - \frac{i(p_n+l)}{p_n(l+k)}\right) \binom{p_n}{i} \binom{p+1-l}{k-i} \\
 &= \sum_i \binom{p_n}{i} \binom{p+1-l}{k-i} - \frac{p_n+l}{l+k} \sum_i \frac{i}{p_n} \binom{p_n}{i} \binom{p+1-l}{k-i} \\
 &= \binom{p'+1-l}{k} - \frac{p_n+l}{l+k} \sum_i \binom{p_n-1}{i-1} \binom{p+1-l}{k-i} \\
 &= \binom{p'+1-l}{k} - \frac{p_n+l}{l+k} \binom{p'-l}{k-1} \\
 &= \binom{p'+1-l}{k} \left(1 - \frac{k(p_n+l)}{(l+k)(p'+1-l)}\right).
 \end{aligned}$$

This completes the proof of the theorem. $\square$

Remark 4.7. The recurrence relation (4.1) for the M -Eulerian–Narayana numbers can be rewritten as

$$A_{M',k} = \sum_{l \leq k} A_{M,l} \left[\binom{p_n+l-1}{k-1} \binom{p'+1-l}{k-l} - \binom{p_n+l}{k} \binom{p'-l}{k-l-1} \right]. \quad (4.9)$$

Since our proof of (4.9) uses heavy computations, it is natural to ask for a more direct combinatorial proof.

5. Concluding remarks, open problems

The main achievement of this paper is a combinatorial interpretation in terms of weakly increasing trees for the γ -coefficients of the p -multiset Eulerian polynomials. This is accomplished by proving an equidistribution (see [Theorem 3.4](#)) of internal nodes over multiset trees and weakly increasing trees with the aid of Chen's general bijective algorithm for trees.

The class of weakly increasing trees on a multiset has been found rich of combinatorial connections (see [\[14,15\]](#)) since the introduction of it in [\[15\]](#). One elementary result proved in [\[15\]](#) is the compact product formula

$$|\mathcal{T}_M| = \frac{1}{1 + N_n} \prod_{i=1}^n \binom{N_i + p_i}{p_i},$$

where $M = \{1^{p_1}, \dots, n^{p_n}\}$ and $N_i = p_1 + \dots + p_i$. As a specialization of [Corollary 4.3](#), we have

$$|\mathcal{T}_M| = |\mathcal{T}_{M'}| \quad \text{with } M = \{1^{2n+2}, 2^n\} \text{ and } M' = \{1^{2n+1}, 2^{n+1}\}.$$

Surprisingly, if we restrict the multisets to have only two different letters, then there seems to be no more equinumerosity other than the above ones. This suspicion is equivalent to the following Diophantine equation conjecture, as

$$|\mathcal{T}_{\{1^x, 2^y\}}| = \frac{1}{1 + x + y} \binom{2x}{x} \binom{x+2y}{y}.$$

Conjecture 5.1. Let $f(x, y) := \frac{1}{1+x+y} \binom{2x}{x} \binom{x+2y}{y}$. The solutions for the Diophantine equation $f(x, y) = f(x', y')$, where $x, y, x', y' \in \mathbb{P}$ and $x \neq x'$, are $(x, y) = (2n + 1, n + 1)$ and $(x', y') = (2n + 2, n)$ for all $n \in \mathbb{P}$.

This conjecture has been verified by computer for $x, y, x', y' \leq 500$. A more challenging open problem would be:

Problem 5.2. Characterize all multisets M and M' for which $|\mathcal{T}_M| = |\mathcal{T}_{M'}|$.

Another instance for [Problem 5.2](#) is $M = \{1^3, 2^2\}$ and $M' = \{1^1, 2^3, 3^1\}$, which indicates that M and M' may have different number of distinct letters.

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